Relative-fit and nested difference tests for estimated-weight moment-quadratic estimators

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Purpose

The estimated-weight sandwich (companion note `weighted_moment_ij_grid`) corrects the parameter covariance by adding a weight-derivative term. This note asks whether the same correction extends to model-comparison tests, and whether the nested difference test follows as a corollary. Three answers, stated up front.

1. The correction is proportional to the pseudo-true residual r_* (Proposition 1). It is identically zero for the goodness-of-fit null, where the focal model is assumed correct, and present for every test whose null does not assume correctness.

2. For distinguishable models the relative-fit (Vuong) variance is a clean corollary: $\omega^2 = c^\mathsf{T}\Gamma c$ with the weight channel carried by an explicit term in c (Proposition 2).

3. The nested, degenerate case is *not* a clean corollary. The weight channel and the bread curvature both deform Satorra's projection, so the naive substitution $\Gamma \mapsto (W - W_r)\Gamma(W - W_r)^\mathsf{T}$ into the U-matrix is wrong (Proposition 3). It needs its own derivation.

1 Recap of the corrected influence

The estimator minimises $F_N(\theta) = \frac{1}{2}r(\theta)^\mathsf{T}\widehat{W}r(\theta)$ with residual $r(\theta) = \eta(\theta) - u$, Jacobian $\Delta = \partial\eta/\partial\theta^\mathsf{T}$, and weight $\widehat{W} = W(u)$ built from the sample moments u. Write g_i for the moment influence function, $\Gamma = \mathbb{E}[g_i g_i^\mathsf{T}]$, and

θ_*, $r_* = \eta(\theta_*) - u_0$ for the pseudo-true value and residual. The estimator is asymptotically linear with influence function $\phi_i = H^{-1}\Delta^\top v_i$, where

$$H = A + C, \quad A = \Delta^\top W \Delta, \quad C = \sum_k (W r_*)_k \nabla^2 \eta_k(\theta_*), \quad v_i = W g_i - \dot{W}_i\, r_*, \tag{1}$$

where $\dot{W}_i$ is the casewise influence of the weight. When the weight is a function of the modelled moments u, $\dot{W}_i r_* = W_r g_i$ with $W_r = [\,\partial W/\partial u_1\, r_*,\ \ldots,\ \partial W/\partial u_m\, r_*\,]$ the residual-contracted weight derivative, so $v_i = (W - W_r) g_i$ and the corrected meat is $\Omega = (W - W_r)\Gamma(W - W_r)^\top$. In general $\Omega = \mathrm{Var}(v_i)$ is formed from the casewise rows; the matrix form is the special case used in Sections 3–5.

2 The residual governs the correction

Proposition 1. *Every estimated-weight correction in* (1)*, namely the curvature term C and the weight term W_r, is linear in the pseudo-true residual r_*. Hence both vanish exactly when $r_* = 0$. For a test whose null fixes the focal model as correctly specified the correction is identically zero; for a test whose null leaves the focal model misspecified the correction is present.*

Proof. C contracts $W r_*$ and W_r contracts r_* columnwise, so both are $O(\|r_*\|)$ and vanish at $r_* = 0$. The goodness-of-fit null sets $\eta(\theta_*) = u_0$, hence $r_* = 0$. The relative-fit and the not-assuming-correct nested nulls leave $r_* \neq 0$. $\square$

A second, sharper reason closes the goodness-of-fit case. At that null the fitted residual is $r(\widehat{\theta}) = O_p(N^{-1/2})$, so the weight fluctuation enters the residual only through $(\widehat{W} - W) r(\widehat{\theta}) = O_p(N^{-1})$, which is negligible. The estimated weight is therefore asymptotically irrelevant to the fit statistic, which is why the fixed-weight projection theory of [1] and the scaled statistic implemented in practice are already correct under the null.

3 Goodness of fit: the projection, unchanged

For a fixed weight, expand the fitted residual at the goodness-of-fit null. With $H = A$ and $W_r = 0$ there,

$$\sqrt{N}\, r(\widehat{\theta}) = -M\sqrt{N}(u - u_0) + o_p(1), \qquad M = I - \Delta A^{-1}\Delta^\top W,$$

so $\mathrm{Var}(\sqrt{N}\, r(\widehat{\theta})) = M\Gamma M^\top$ and the fit statistic $N\, r^\top W r$ converges to $\sum_j \lambda_j \chi^2_{1,j}$ with $\lambda_j = \mathrm{eig}(U\Gamma)$ and $U = WM = W - W\Delta(\Delta^\top W \Delta)^{-1}\Delta^\top W$ [1]. When

$W = \Gamma^{-1}$ the product $U\Gamma$ is idempotent and the law collapses to χ^2_{df}; for $W \neq \Gamma^{-1}$ (DWLS, ULS) it is a genuine mixture, the reason the scaled statistic [3] exists. By Proposition 1 the estimated weight contributes nothing here.

4 Relative fit: a clean corollary

Compare two structural models A and B on the same indicators. The weight $\widehat{W} = W(u)$ is a function of the sample moments alone, so it is shared, $W_A = W_B = W$. Let $D = F_A(\widehat{\theta}_A) - F_B(\widehat{\theta}_B)$ be the sample discrepancy difference and r_*^A, r_*^B the two pseudo-true residuals.

Proposition 2. *The discrepancy difference is asymptotically linear,* $\sqrt{N}\,D = N^{-1/2}\sum_i c^{\mathsf{T}} g_i + o_p(1)$, *with*

$$c = -W(r_*^A - r_*^B) + \tfrac{1}{2}(b^A - b^B), \qquad b_l^A = (r_*^A)^{\mathsf{T}} \frac{\partial W}{\partial u_l}\, r_*^A, \tag{2}$$

and b_l^B *defined likewise. For distinguishable models* $\omega^2 = c^{\mathsf{T}}\Gamma c > 0$ *and* $\sqrt{N}\,D/\widehat{\omega} \to \mathcal{N}(0,1)$.

Proof. View F_A as a functional of the sampling distribution through u, the minimiser $\widehat{\theta}_A(u)$, and the weight $W(u)$. The minimiser is interior, so the envelope theorem removes the $\widehat{\theta}_A$ channel and

$$\mathrm{IF}_i(F_A) = \Big(\tfrac{\partial F_A}{\partial u} + \tfrac{\partial F_A}{\partial W}\, DW(u_0)\Big) g_i = \big(-W r_*^A + \tfrac{1}{2} b^A\big)^{\mathsf{T}} g_i,$$

using $\partial F_A/\partial u = -r^{\mathsf{T}} W$ and $\partial(\frac{1}{2} r^{\mathsf{T}} W r)/\partial W$ contracted with $DW[g_i] = \sum_l (\partial W/\partial u_l) g_{i,l}$. Subtracting $\mathrm{IF}_i(F_B)$ gives (2); the central limit theorem gives the normal law. □

The term $-W(r_*^A - r_*^B)$ is the difference of weighted residuals that a fixed-weight account would give. The term $\frac{1}{2}(b^A - b^B)$ is the weight channel, quadratic in each model's own residual and contracted with the weight derivative. It is the testing analogue of W_r and is the new ingredient. Every quantity is already available: r_*^A, r_*^B from the two fits, $\partial W/\partial u$ from the same derivative the standard error uses, and Γ from the stage-one moments. The relative-fit test is therefore a corollary of the standard-error machinery, with no projection involved because the envelope theorem has removed the parameter channel.

5 The nested case does not commute

When B is nested in A the discrepancy difference degenerates under the null that the restriction holds, $r_*^A = r_*^B$ in the relevant directions, so $c = 0$ and $\omega^2 = 0$. The first-order term vanishes and the limiting law is the second-order quadratic form, which is where the projection returns. The question is whether that quadratic form is Satorra's U-difference with the meat Γ simply replaced by the corrected Ω.

Proposition 3. *Off the goodness-of-fit null the fitted residual fluctuation is*

$$\sqrt{N}\,(r_A(\widehat{\theta}_A) - r_*^A) = -\widetilde{M}_A\sqrt{N}(u-u_0) + o_p(1), \qquad \widetilde{M}_A = I - \Delta_A H_A^{-1}\Delta_A^\mathsf{T}(W - W_r^A),$$

with $H_A = A_A + C_A$. The nested difference statistic converges to $\sum_j \lambda_j \chi_{1,j}^2$ with $\lambda_j = \mathrm{eig}\big[(\widetilde{M}_A^\mathsf{T} W \widetilde{M}_A - \widetilde{M}_B^\mathsf{T} W \widetilde{M}_B)\Gamma\big]$. The generalised annihilator $\widetilde{M}_A$ carries both the bread curvature C_A and the weight channel W_r^A inside the projector, so it does not equal the fixed-weight annihilator $M_A = I - \Delta_A A_A^{-1}\Delta_A^\mathsf{T} W$. Consequently the law is not $\mathrm{eig}[(U_B - U_A)\Omega]$*, and the substitution $\Gamma \mapsto \Omega$ into the U-matrix is incorrect.*

Proof. Expand $r_A(\widehat{\theta}_A) = r_*^A + \Delta_A(\widehat{\theta}_A - \theta_*^A) - (u - u_0) + o_p(N^{-1/2})$ and substitute the corrected influence $\sqrt{N}(\widehat{\theta}_A - \theta_*^A) = H_A^{-1}\Delta_A^\mathsf{T}(W - W_r^A)\sqrt{N}(u - u_0) + o_p(1)$ from (1). Collecting terms gives the stated $\widetilde{M}_A$. The second-order term of $N(F_B - F_A)$ is the difference of the quadratic forms $z^\mathsf{T}\widetilde{M}_A^\mathsf{T} W \widetilde{M}_A z$ and $z^\mathsf{T}\widetilde{M}_B^\mathsf{T} W \widetilde{M}_B z$ with $z \sim \mathcal{N}(0, \Gamma)$, whence the eigenvalue characterisation. The fixed-weight reduction sets $C_A = 0$ and $W_r^A = 0$, returning $\widetilde{M}_A = M_A$ and $\lambda_j = \mathrm{eig}[(U_B - U_A)\Gamma]$ of [2]. □

The correction does not factor out of the projection. The weight channel enters the same place as the bread curvature, inside the annihilator, so the meat and the projector are entangled. Computing the nested reference law requires the generalised annihilators $\widetilde{M}$ rather than a meat swap, and that is a separate derivation, not a corollary.

6 Specialization: complete-data DWLS on a linear covariance model

The curvature term $C = \sum_k (W r_*)_k \nabla^2 \eta_k(\theta_*)$ is the only place the shape of the model surface enters. It vanishes whenever η is linear in θ, which removes one of the two deformations and isolates the weight channel.

6.1 A model with $C = 0$

Take a linear covariance structure $\Sigma(\theta) = \sum_k \theta_k G_k$ with fixed symmetric G_k, so $\eta(\theta) = \operatorname{vech}\Sigma(\theta) = \Delta\theta$ with $\Delta = [\operatorname{vech} G_1, \dots, \operatorname{vech} G_q]$ constant, $\nabla^2\eta_k = 0$, hence $C = 0$ and $H = A = \Delta^{\mathsf{T}} W \Delta$ constant. A concrete nested pair on p continuous variables: B is compound symmetry (one variance, one covariance), nested in A (one variance, free covariances), the difference testing equality of the covariances. Both are linear, so the only surviving deformation is the weight channel.

6.2 DWLS ingredients

The stage-one moments are $u = \operatorname{vech}(S)$ with casewise influence $g_i = \operatorname{vech}(z_i z_i^{\mathsf{T}}) - \sigma_0$, $z_i = y_i - \mu$, and $\Gamma = \mathbb{E}[g_i g_i^{\mathsf{T}}]$ the Browne fourth-moment matrix. DWLS uses $\widehat{W} = \operatorname{diag}(\widehat{\Gamma})^{-1}$, so with $\gamma = \operatorname{diag}(\Gamma)$ the population weight is $W = \operatorname{diag}(\gamma)^{-1}$. The weight influence acts diagonally,

$$\dot{w}_{i,k} = -\gamma_k^{-2}\,\dot{\gamma}_{i,k}, \qquad \dot{\gamma}_{i,k} = g_{i,k}^2 - \gamma_k + (\text{centring corrections}),$$

where $\dot{\gamma}_{i,k}$ is the casewise influence of the k-th diagonal of $\widehat{\Gamma}$. The corrected casewise moment-space influence is

$$v_{i,k} = \frac{g_{i,k}}{\gamma_k} + \frac{r_{*,k}\,\dot{\gamma}_{i,k}}{\gamma_k^2}, \tag{3}$$

which matches the implemented DWLS adapter.

Remark (The DWLS weight channel is not a matrix on g_i). $\widehat{W} = \operatorname{diag}(\widehat{\Gamma})^{-1}$ depends on fourth-order sample moments, not on the modelled $u = \operatorname{vech}(S)$. Its influence $\dot{\gamma}_{i,k}$ is quadratic in g_i (through $g_{i,k}^2$), so the weight channel in (3) is a genuine casewise quantity, not $W_r g_i$ for a matrix W_r. The matrix annihilator $\widetilde{M}$ of Proposition 3 therefore holds only when the weight is a function of u; for DWLS the residual influence $\rho_i = \Delta A^{-1}\Delta^{\mathsf{T}} v_i - g_i$ stays casewise and carries a part quadratic in g_i, so its covariance reaches eighth-order moments of the data.

6.3 The verification-friendly twin: normal-theory weight

The normal-theory weight $W = W_{\mathrm{NT}}(S)$ is a function of u, so it is the clean case. With $C = 0$ and a data-estimated weight, W_r is a constant matrix, $\widetilde{M} = I - \Delta A^{-1}\Delta^{\mathsf{T}}(W - W_r)$ is constant, and the nested law is

$$\lambda_j = \operatorname{eig}\big[(\widetilde{M}_B^{\mathsf{T}} W \widetilde{M}_B - \widetilde{M}_A^{\mathsf{T}} W \widetilde{M}_A)\Gamma\big],$$

all constant matrices. A fixed weight would give $W_r = 0$ and, with $C = 0$, the undeformed Satorra law, so the data-estimated weight is exactly what produces the phenomenon. This isolates the weight channel from the curvature and is the easiest case to verify by hand and to eigendecompose.

6.4 The geometry of the payoff

The deformation is $\widetilde{M} - M = \Delta A^{-1}\Delta^{\mathsf{T}} W_r$, linear in r_* and zero at $r_* = 0$. The nested statistic sees r_* only through the rank-$(\mathrm{df}_B - \mathrm{df}_A)$ map $\widetilde{M}_B - \widetilde{M}_A$ onto the constraint directions. The exploration is to sweep the direction of r_* at fixed $\|r_*\|$: generate from $\Sigma_0 = \Sigma_B(\theta_*) + \epsilon D$ and rotate D from W-orthogonal to the constraint space toward aligned. The conjecture, which the simulation tests rather than assumes, is that orthogonal r_* leaves the law near Satorra (the benign regime where robust intervals already calibrate) and aligned r_* deforms it most. Because the deformation is analytic in r_*, the predicted miscalibration can overlay the simulated rejection rate.

7 Verdict

The testing extension partitions cleanly. The goodness-of-fit test needs no correction (Proposition 1); the prevailing scaled statistic is already right. The relative-fit test for distinguishable models is a corollary of the standard-error machinery, with $\omega^2 = c^{\mathsf{T}}\Gamma c$ and the weight channel (2) computable from quantities the standard error already forms (Proposition 2). The nested difference test, in the regime that does not assume the larger model correct, is its own project, because the weight channel deforms Satorra's projection rather than its meat (Proposition 3). For the manuscript this argues to keep the estimated-weight standard error and the distinguishable relative-fit test together, and to treat the nested weighted-χ^2 law as a sequel [4, 5].

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